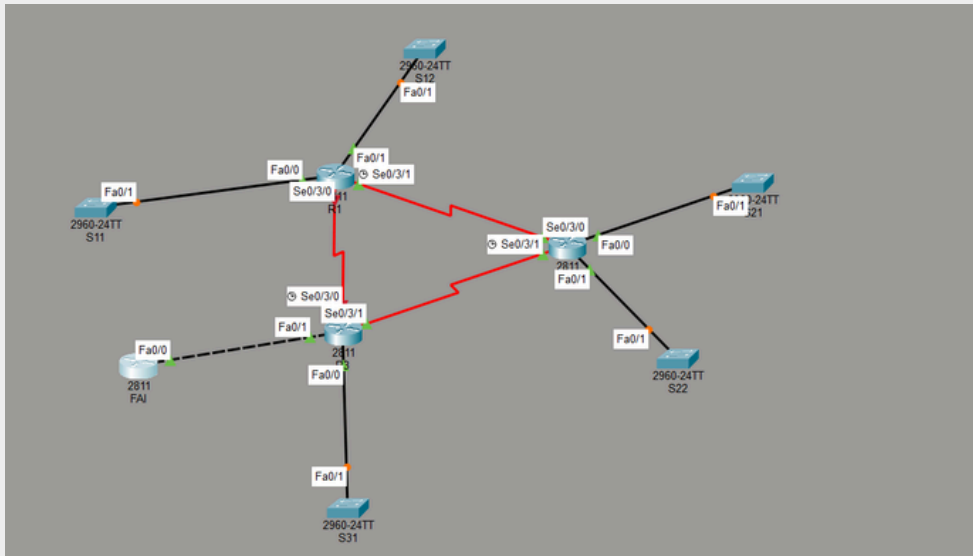


Introduction

L'objectif général de ce Lab est de comprendre et maîtriser la configuration du routage statique pour assurer la connectivité réseau entre plusieurs sous-réseaux utilisant les protocoles IPv4 et IPv6, tout en garantissant l'efficacité, la sécurité et la fiabilité du réseau.



1. Configurez les paramètres de base du routeur R1

1.a

```
Router>enable
Router#conf t
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#ip domain-name fsm.local
R1(config)#no ip domain-lookup
R1(config)#service password-encryption
R1(config)#banner motd # Les acces sans autorisation sont strictement interdits #
R1(config)#enable secret class
R1(config)#username admin secret chu
R1(config)#line console 0
R1(config-line)#password fsm
R1(config-line)#exec-timeout 0 0
R1(config-line)#login
```

1.b

```
R1(config-line)#crypto key generate rsa
The name for the keys will be: R1.fsm.local
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R1(config)#ip ssh version 2
*Mar 1 0:26:39.395: %SSH-5-ENABLED: SSH 1.99 has been enabled
R1(config)#line vty 0 15
R1(config-line)#login local
R1(config-line)#transport input ssh
R1(config-line)#logging synchronous
R1(config-line)#exit
```

1.c

```
R1(config)#interface F0/0
R1(config-if)#description "LAN vers S11"
R1(config-if)#ip address 192.168.0.126 255.255.255.128
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R1(config-if)#interface F0/1
R1(config-if)#description "LAN vers S12"
R1(config-if)#ip address 192.168.0.254 255.255.255.128
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

R1(config-if)#interface S0/3/0
R1(config-if)#description "WAN vers R2 (DCE)"
R1(config-if)#ip address 172.16.23.1 255.255.255.252
R1(config-if)#clock rate 250000
This command applies only to DCE interfaces
R1(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
R1(config-if)#interface S0/3/1
R1(config-if)#description "WAN vers R3 (DTE)"
R1(config-if)#ip address 172.16.41.2 255.255.255.252
R1(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/1, changed state to down
...
```

2. Configurez les paramètres de base du routeur R2 :

2.a

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R2
R2(config)#ip domain-name fsm.local
R2(config)#no ip domain-lookup
...
```

2.b

```
R2(config)#interface F0/0
R2(config-if)#description "LAN vers S21"
R2(config-if)#ip address 192.168.1.254 255.255.255.0
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R2(config-if)#interface F0/1
R2(config-if)#description "LAN vers S22"
R2(config-if)#ip address 192.168.2.254 255.255.255.0
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

R2(config-if)#interface S0/3/0
R2(config-if)#description "WAN vers R3 (DCE)"
R2(config-if)#ip address 172.16.34.1 255.255.255.252
R2(config-if)#clock rate 250000
This command applies only to DCE interfaces
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

R2(config-if)#interface S0/3/1
R2(config-if)#description "WAN vers R1 (DTE)"
R2(config-if)#ip address 172.16.23.2 255.255.255.252
R2(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/1, changed state to down
R2(config-if)#exit
R2(config)#exit
R2#
%SYS-5-CONFIG_I: Configured from console by console
```

```
R2#ping 172.16.23.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.23.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

R2#sh ip route | begin Gateway
^
% Invalid input detected at '^' marker.

R2#sh ip route
^
% Invalid input detected at '^' marker.

R2#sh ip route | begin Gateway
Gateway of last resort is not set

    172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.34.0/30 is directly connected, Serial0/3/0
L       172.16.34.1/32 is directly connected, Serial0/3/0
C       192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, FastEthernet0/0
L       192.168.1.254/32 is directly connected, FastEthernet0/0
C       192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, FastEthernet0/1
L       192.168.2.254/32 is directly connected, FastEthernet0/1

R2#
```

3. Configurez les paramètres de base du routeur R3 :

3.a

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R3
R3(config)#ip domain-name fsm.local
R3(config)#no ip domain-lookup
```

3.b

```
R3(config)#interface F0/0
R3(config-if)#description "LAN vers S31"
R3(config-if)#ip address 192.168.3.254 255.255.255.0
R3(config-if)#no shutdown

R3(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R3(config-if)#interface F0/1
R3(config-if)#description "Vers FAI"
R3(config-if)#ip address dhcp
R3(config-if)#no shutdown

R3(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

R3(config-if)#interface S0/3/0
R3(config-if)#description "WAN vers R1 (DCE)"
R3(config-if)#ip address 172.16.41.1 255.255.255.252
R3(config-if)#clock rate 64000
R3(config-if)#no shutdown
^
% Invalid input detected at '^' marker.

R3(config-if)#no shutdown

R3(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

R3(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

R3(config-if)#interface S0/3/1
R3(config-if)#description "WAN vers R2 (DTE)"
R3(config-if)#ip address 172.16.34.2 255.255.255.252
R3(config-if)#no shutdown

R3(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/1, changed state to up
```

4.b

4. Configurez le routeur du FAI : utilisez la configuration ci-dessous.

4.b

```
R3> sh ip inter f0/1

FastEthernet0/1 is up, line protocol is up (connected)
Internet address is 200.0.0.51/24
Broadcast address is 255.255.255.255
Address determined by DHCP
```

5. Configurez les routes statiques sur le routeur R3 :

5.a.b

```
R3>enable
R3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#ipv6 route ::/0 2001:db8:acad:200::1
R3(config)#ipv6 route 2001:db8:acad:1::/64 2001:db8:acad:34::1
R3(config)#ipv6 route 2001:db8:acad:2::/64 2001:db8:acad:34::1
R3(config)#ipv6 route 2001:db8:acad:3::/64 2001:db8:acad:34::1
R3(config)#ipv6 route 2001:db8:acad:4::/64 2001:db8:acad:34::1
R3(config)#ipv6 route 2001:db8:acad:23::/64 2001:db8:acad:34::1
R3(config)#
```

6.

```
R2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#ipv6 route 2001:db8:acad:1::/64 2001:db8:acad:23::1
R2(config)#ipv6 route 2001:db8:acad:2::/64 2001:db8:acad:23::1
R2(config)#ipv6 route 2001:db8:acad:16::/64 2001:db8:acad:23::1
R2(config)#ipv6 route 2001:db8:acad:16::/64 2001:db8:acad:23::2
R2(config)#ipv6 route ::/0 2001:db8:acad:34::2
R2(config)#
```

7.

```
R1>enable
Password:
Password:
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)# ip route 0.0.0.0 0.0.0.0 172.16.23.2
R1(config)#
```

8.

```
R1(config)#ip route 0.0.0.0 0.0.0.0 172.16.41.1 240
R1(config)# inter s0/3/0
R1(config-if)#shut
R1(config-if)# inter s0/0/0
%Invalid interface type and number
R1(config)# inter s0/3/0
R1(config-if)#shut
R1(config-if)# inter Se0/3/0
R1(config-if)#shut
R1(config-if)# inter Se0/3/1
R1(config-if)#shut

R1(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/1, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/1, changed state to down
```

9.

```
R3>enable
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#ip route 192.168.0.0 255.255.252.0 172.16.41.2 240
R3(config)# ip route 172.16.23.0 255.255.255.252 172.16.41.2 240
R3(config)# inter s0/3/0
R3(config-if)#shut

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to administratively down
R3(config-if)#do traceroute 192.168.0.254
Type escape sequence to abort.
Tracing the route to 192.168.0.254

 0  *    *    *
 1  *    *    *
 2  *    *    *
 3  *    *    *
 4  *    *    *
 5  *    *    *
 6  *    *    *
 7  *    *    *
```

1.c

```
R1(config)#interface F0/0
R1(config-if)#description "LAN vers S11"
R1(config-if)#ipv6 address 2001:db8:acad:1::254/64
R1(config-if)#no shutdown
R1(config-if)#interface F0/1
R1(config-if)#description "LAN vers S12"
R1(config-if)#ipv6 address 2001:db8:acad:2::254/64
R1(config-if)#no shutdown
R1(config-if)#interface S0/0/0
%Invalid interface type and number
R1(config)#interface S0/3/0
R1(config-if)#description "WAN vers R2 (DCE)"
R1(config-if)#ipv6 address 2001:db8:acad:23::1/64
R1(config-if)# clock rate 250000
This command applies only to DCE interfaces
R1(config-if)# no shutdown

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
R1(config-if)#interface S0/3/1
R1(config-if)#description "WAN vers R3 (DTE)"
R1(config-if)#ipv6 address 2001:db8:acad:41::2/64
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/1, changed state to up
```

2.b

```
R2>enable
R2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#interface F0/0
R2(config-if)#description "LAN vers S21"
R2(config-if)# ipv6 address 2001:db8:acad:3::254/64
R2(config-if)#no shutdown
R2(config-if)#interface F0/1
R2(config-if)# description "LAN vers ^Z
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#description "LAN vers S22"
^
% Invalid input detected at '^' marker.

R2(config)#interface F0/1
R2(config-if)#description "LAN vers S22"
R2(config-if)#ipv6 address 2001:db8:acad:4::254/64
R2(config-if)#no shutdown
R2(config-if)#interface S0/3/0
R2(config-if)#description "WAN vers R3 (DCE)"
R2(config-if)#description "WAN vers R3 (DCE)"
R2(config-if)#ipv6 address 2001:db8:acad:34::1/64
R2(config-if)#clock rate 250000
This command applies only to DCE interfaces
R2(config-if)#no shutdown
R2(config-if)#interface S0/0/1
%Invalid interface type and number
R2(config)#interface S0/3/1
R2(config-if)#description "WAN vers R1 (DTE)"
R2(config-if)#ipv6 address 2001:db8:acad:23::2/64
R2(config-if)#no shutdown
R2(config-if)#ipv6 unicast-routing
R2(config)#
```

```
R2#sh ipv6 route
IPv6 Routing Table - 13 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
S    ::/0 [1/0]
    via 2001:DB8:ACAD:34::2
S    2001:DB8:ACAD:1::/64 [1/0]
    via 2001:DB8:ACAD:23::1
S    2001:DB8:ACAD:2::/64 [1/0]
    via 2001:DB8:ACAD:23::1
C    2001:DB8:ACAD:3::/64 [0/0]
    via FastEthernet0/0, directly connected
L    2001:DB8:ACAD:3::254/128 [0/0]
    via FastEthernet0/0, receive
C    2001:DB8:ACAD:4::/64 [0/0]
    via FastEthernet0/1, directly connected
L    2001:DB8:ACAD:4::254/128 [0/0]
    via FastEthernet0/1, receive

R2#sh ipv6 inter br
FastEthernet0/0                [up/up]
    FE80::230:F2FF:FE88:66A1
    2001:DB8:ACAD:3::254
FastEthernet0/1                [up/up]
    FE80::209:7CFF:FE53:DC44
    2001:DB8:ACAD:4::254
Serial0/3/0                    [up/up]
    FE80::230:F2FF:FE88:66A1
    2001:DB8:ACAD:34::1
Serial0/3/1                    [up/up]
    FE80::230:F2FF:FE88:66A1
    2001:DB8:ACAD:23::2
Vlan1                          [administratively down/down]
    unassigned
--
```

3.b

```
R3>enable
R3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#interface F0/0
R3(config-if)#description "LAN vers S31"
R3(config-if)#ipv6 address 2001:db8:acad:16::254/64
R3(config-if)#no shutdown
R3(config-if)#interface F0/1
R3(config-if)#description "Vers FAI"
R3(config-if)#ipv6 address 2001:db8:acad:200::2/64
R3(config-if)#no shutdown
R3(config-if)#interface S0/3/0
R3(config-if)#description "WAN vers R1 (DCE)"
R3(config-if)#ipv6 address 2001:db8:acad:41::1/64
R3(config-if)#clock rate 64000
R3(config-if)#no shutdown

R3(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

R3(config-if)#interface S0/3/1
R3(config-if)#description "WAN vers R2 (DTE)"
R3(config-if)#ipv6 address 2001:db8:acad:34::2/64
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#ipv6 unicast-routing
```

```
R3>enable
R3#sh ipv6 inter br
FastEthernet0/0          [up/up]
    FE80::240:BFF:FEA8:3080
    2001:DB8:ACAD:16::254
FastEthernet0/1          [up/up]
    FE80::201:96FF:FEE1:7111
    2001:DB8:ACAD:200::2
Serial0/3/0              [up/up]
    FE80::240:BFF:FEA8:3080
    2001:DB8:ACAD:41::1
Serial0/3/1              [up/up]
    FE80::240:BFF:FEA8:3080
    2001:DB8:ACAD:34::2
Vlan1                    [administratively down/down]
    unassigned
R3#
```

5.a

```
R3(config)#ipv6 route ::/0 2001:db8:acad:200::1
```

5.b

```
R3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#ipv6 route ::/0 2001:db8:acad:200::1
R3(config)#ipv6 route 2001:db8:acad:1::/64 2001:db8:acad:34::1
R3(config)#ipv6 route 2001:db8:acad:2::/64 2001:db8:acad:34::1
R3(config)#ipv6 route 2001:db8:acad:3::/64 2001:db8:acad:34::1
R3(config)#ipv6 route 2001:db8:acad:4::/64 2001:db8:acad:34::1
R3(config)#
```

6

```
R2(config)#ipv6 route 2001:db8:acad:1::/64 2001:db8:acad:23::1
R2(config)#ipv6 route 2001:db8:acad:2::/64 2001:db8:acad:23::1
R2(config)#ipv6 route 2001:db8:acad:16::/64 2001:db8:acad:34::2
R2(config)#ipv6 route ::/0 2001:db8:acad:34::2
```

7

```
R1(config)#ipv6 route ::/0 2001:db8:acad:23::2
```

8

```
R1(config)#ipv6 route ::/0 2001:db8:acad:41::1 240
R1(config)#inter s0/3/0
R1(config-if)#shut

R1(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to down
```

9

```
R3(config)#ipv6 route 2001:db8:acad:0::/61 2001:db8:acad:41::2 240
R3(config)#ipv6 route 2001:db8:acad:23::/64 2001:db8:acad:41::2 240
```

conclusion :

Ce TP a permis de mettre en pratique la configuration complète du routage statique sur des routeurs Cisco, aussi bien en IPv4 qu'en IPv6. Les principales compétences acquises sont :

- Configuration des interfaces : attribution d'adresses IPv4 et IPv6, activation des liaisons série avec clock rate, configuration DHCP client.
- Routes statiques : mise en place de routes spécifiques et de routes par défaut (0.0.0.0/0 ou ::/0) en choisissant le chemin optimal selon la bande passante.
- Routes résumées : optimisation du nombre d'entrées dans la table de routage via la supernetting (ex. 192.168.0.0/22 ou 2001:db8:acad:0::/61).
- Routes flottantes : mise en place de mécanismes de redondance avec une distance administrative élevée (AD=240) pour basculer automatiquement en cas de panne.
- Limites du routage statique : absence de convergence automatique les routeurs non directement affectés par une panne ne la détectent pas, ce qui crée des boucles ou des routes invalides.